STANDATA interpretation 23-FCI-004/23-BCI-009

Fire/Building

Fixed fire suppression and exhaust systems – cooking, cleaning, maintenance and qualified persons

Date Issued: December 2024

Purpose

This clarifies the requirements from the National Building Code – 2023 Alberta Edition (NBC(AE)) and National Fire Code – 2023 Alberta Edition (NFC(AE)) for fixed fire suppression and exhaust systems for commercial cooking.

Discussion

Owners, designers, and safety code officers (SCO) have asked Municipal Affairs for guidance regarding fixed fire suppression and exhaust systems of non-residential cooking installations and the cleaning thereof. The following responses are intended to address the most raised questions in applying both the NBC(AE) and NFC(AE) for fixed fire suppression and exhaust systems for cooking as well as the requirements for cleaning.

Application

There are standards that apply to fixed fire suppression and exhaust systems in commercial kitchens.

The **installation** of fixed fire suppression and exhaust systems is under the jurisdiction of the NBC(AE). The authority having jurisdiction (AHJ) in the building discipline must be consulted for building permit and professional involvement requirements.

The **use, inspection, testing, cleaning** and **maintenance** of fixed fire suppression and exhaust systems are under the jurisdiction of the NFC(AE). As such, the AHJ in the fire discipline should be consulted with regards to the use, inspection, testing and maintenance of these systems.

The NBC(AE) and the NFC(AE) sets out the minimum requirements for fixed fire suppression and exhaust systems. Through these two codes, standards in the National Fire Protection Association (NFPA) documents are referenced, including:

1. NFPA 96 “Standard for Ventilation Control and Fire Protection of Commercial Cooking Operations”
2. NFPA 10 “Standard for Portable Fire Extinguishers”

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NBC(AE) Division B

3.3.1.2.(2) Hazardous Substances, Equipment and Processes
1) Systems for the ventilation of cooking equipment that is not within a *dwelling unit* and is used in processes producing grease-laden vapours shall be designed and installed in conformance with Articles 3.6.3.5., 6.3.1.6. and 6.9.1.3. (See Note A-3.3.1.2.(2).)

A-3.3.1.2.(2) Cooking Equipment Ventilation. Cooking equipment manufactured for use in dwelling units and other residential suites is often installed in buildings used for assembly and care, treatment or detention purposes. It is not obvious from the Code requirements or those of NFPA 96, "Standard for Ventilation Control and Fire Protection of Commercial Cooking Operations," whether a ventilation and grease removal system is required in all assembly and care, treatment or detention uses. If the equipment is to be used in a manner that will produce grease-laden vapours that are substantially more than would be produced in a normal household environment, then it would be appropriate to apply the requirements of NFPA 96. If the equipment is used primarily for reheating food prepared elsewhere or is used occasionally for demonstration or educational purposes, there would be no expectation of applying the requirements of NFPA 96. In all cases the circumstances should be reviewed with the authority having jurisdiction.

9.10.1.4. Items under Part 6 Jurisdiction
1) In kitchens containing commercial cooking equipment used in processes producing grease-laden vapours, the equipment shall be designed and installed in conformance with Article 6.3.1.6. (See Note A-9.10.1.4.(1).)

A-9.10.1.4.(1) Commercial Cooking Equipment. Part 6 refers to NFPA 96, "Standard for Ventilation Control and Fire Protection of Commercial Cooking Operations," which in turn references "Commercial Cooking Equipment." However, the deciding factor as to whether or not NFPA 96 applies is the potential for production of grease-laden vapours and smoke, rather than the type of equipment used. While NFPA 96 does not apply to domestic equipment for normal residential family use, it should apply to domestic equipment used in commercial, industrial, institutional and similar cooking applications where the potential for the production of smoke and grease-laden vapours exceeds that for normal residential family use.

NFC(AE)

2.6.1.9. Commercial Cooking Equipment
1) Commercial cooking equipment exhaust and fire protection systems shall be designed and installed in conformance with the NBC(AE).
2) Except as required in Sentences (3) to (5), the use, inspection and maintenance of commercial cooking equipment exhaust and fire protection systems shall be in conformance with NFPA 96, "Standard for Ventilation Control and Fire Protection of Commercial Cooking Operations."

A kitchen exhaust system may be required for commercial equipment.

As noted in NBC(AE) Sentence 3.3.1.2.(2), systems for the ventilation of cooking equipment that is used in processes that produce grease-laden vapours must be designed and installed in conformance with Articles 3.6.3.5, 6.3.1.6 and 6.9.1.3.

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Where grease-laden vapours and/or smoke is generated outside an enclosed appliance, the exhaust system must meet the requirements of the NBC(AE) and NFPA 96. Examples of appliances and/or processes that is deemed to create grease-laden vapours are provided in the appendix of NFPA 96.

The hood must meet the requirements of the referenced standards, have welded seamless ductwork and the required labelled access panels. In addition, these systems will require a fire suppression system that complies to the requirements of the NBC(AE), NFC(AE) and NFPA 17A.

NBC(AE) Division B

3.6.3.5. Grease Duct Enclosures
(See Note A-3.6.3.5.)
1) Except as provided in Sentence (2), *fire separations* enclosing grease ducts for commercial cooking operations shall conform to NFPA 96, "Standard for Ventilation Control and Fire Protection of Commercial Cooking Operations."
2) The *fire-resistance rating* of field-applied and factory-built grease duct enclosure assemblies shall be determined in conformance with CAN/ULC-S144, "Standard Method of Fire Resistance Test – Grease Duct Assemblies."

A-3.6.3.5. Grease Duct Enclosures. NFPA 96, "Standard for Ventilation Control and Fire Protection of Commercial Cooking Operations," presents two options for enclosing grease ducts for commercial cooking equipment: the first option is to use continuous fire-rated building component assemblies to enclose the ducts and the second one consists of installing proprietary, fire-rated, field-applied or factory-built grease duct assemblies in accordance with the manufacturer's instructions. These types of enclosure assemblies are evaluated for their resistance to fire and their ability to protect adjacent combustibles through reduced clearances. Although NFPA 96 references other standards that deal with grease duct assemblies, Sentence 3.6.3.5.(2) requires that CAN/ULC-S144, "Standard Method of Fire Resistance Test – Grease Duct Assemblies," be used to determine the fire-resistance rating of factory-built and field-applied grease duct assemblies.

6.3.1.6. Commercial Cooking Equipment
1) Except as provided in Article 3.6.3.5., systems for the ventilation of commercial cooking equipment shall be designed, constructed and installed to conform to NFPA 96, "Standard for Ventilation Control and Fire Protection of Commercial Cooking Operations."

NFPA 96 – Chapter 7 Exhaust Duct Systems

7.1. General

7.1.2. All ducts shall lead directly to the exterior of the building, so as not to unduly increase any fire hazard.

Types of commercial cooking appliance that require a fixed fire suppression system.

While all appliances used in Alberta must be certified for use under the *Safety Codes Act, some* commercial cooking appliances may also be tested to certain standards of fixed fire suppression system protection.

The deciding factor for whether a fixed fire suppression system is required depends on the potential for the cooking appliance to produce grease-laden vapours that exceed normal residential family use. NFPA 96 Sentence 4.1.1.2 specifies the amount of grease discharge that would fall within this scope.

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Tested appliances and devices (hoods and ducts) that require a fixed fire suppression system include:

- Fryers
- Pressure Fryers (with lids)
- Ranges (including residential ranges if they are used in a commercial/ institutional cooking operation, when used to sauté or fry foods)
- Griddles (flat top cast iron cooking surface)
- Open and closed top chain broilers
- Char Broilers (includes gas radiant, electric, lava rock, charcoal, mesquite and wood).
- Woks
- Tilting Skillets
- Braising Pans

Appliances that have not been tested and/or listed but require fixed fire suppression protection based on the volume of grease-laden vapours produced include:

- Induction Cookers when used to sauté or fry
- Horizontal Rotisserie (with no enclosing doors)
- Vertical Rotisserie with the exception for:
    - Equipment that have two or less pre-existing vertical rotisseries used for heating pre-cooked/processed donair/gyro meat. (If any additional vertical rotisseries are added, then a protection system is required for all the rotisseries).
- All new restaurants built to NBC(AE) that have new donair/gyro cookers, and both new and existing shawarma cookers require fixed fire suppression.

When a fixed fire suppression system is required, these standards are applicable.

According to NBC(AE) – 2023, Article 6.9.1.3, fixed fire suppression systems must conform to one of two standards:

1) ANSI/UL 300, “Standard for Fire Testing of Fire Extinguishing Systems for Protection of Commercial Cooking Equipment” or

2) ULC/ORD - C1254.6, “Fire Testing of Restaurant Cooking Area Fire Extinguishing Units”

Notes:

1) Solid fuel fired or supplemented appliances are required to be under a separate exhaust/hood system. This operation is also required to have spark arresters. Refer to NFPA 96 Sentence 14.1.6.

2) Water Wash Canopies with a fire cycle have not been tested to ULC/ORD 1254.6 - 1995 and requires a fire suppression system in addition to the water wash system.

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3) Carbon dioxide (CO2) systems have not been tested to ULC ORD 1254.6 - 1995 and do not meet the requirements of the NBC(AE) or NFC(AE).

4) Sprinkler Systems that have not been tested to ULC ORD 1254.6-1995 do not meet the requirements of the NBC(AE) or NFC(AE) for fixed fire suppression under an exhaust hood.

5) Exhaust Ducts: where there is a damper at the duct interface with the hood, the first duct nozzle must be installed immediately above the damper and an access door for installation and servicing the nozzle(s) must be provided within 18 inches of the damper. The nozzle must not interfere with the operation of the damper – on many occasions the connection between the duct collar and the duct is distorted (during installation) of the duct and the damper will not close.

6) Since the adoption of the Alberta Building Code 1997 (ABC 1997), where there is a kitchen fixed fire suppression system installed in a building with a fire alarm system, the fixed fire suppression system must be tied into the building fire alarm system so that, if there is a system discharge, the building fire alarm must activate.

NBC(AE) Division B

6.9.1.3. Commercial Cooking Equipment
1) Fire protection systems for commercial cooking equipment referred to in Sentence 6.3.1.6.(1) using vegetable oil or animal fat shall conform to
a) ANSI/CAN/UL/ULC 300, “Standard for Fire Testing of Fire Extinguishing Systems for Protection of Commercial Cooking Equipment,” or
b) ULC/ORD-C1254.6, “Fire Testing of Restaurant Cooking Area Fire Extinguishing System Units.”

NFPA 96

14.1.6 Solid fuel cooking operations shall have spark arresters to minimize the passage of airborne sparks and embers into plenums and ducts.

A portable fire extinguisher is required in commercial kitchens.

As required by NFPA 96 and NFPA 10, a K Class portable extinguisher is required for the protection of cooking appliances that use combustible cooking media (vegetable or animal oils and fats). These need to be installed within 9.1m of the appliances. If the cooking appliances are protected by an automatic fixed fire suppression system, the portable extinguishers act as a secondary backup source only. This needs to be identified by a placard placed near the fire extinguisher.

Notes:

1) A portable fire extinguisher is required in a commercial kitchen regardless of a fixed fire suppression system installed for the cooking appliance.

2) Type B gas type fire extinguishers are not permitted in a kitchen.

NFPA 96

10.9.1 Portable fire extinguishers shall be selected and installed in kitchen cooking areas in accordance with NFPA 10 and shall be specifically listed for such use.

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10.9.2 Class K fire extinguishers shall be provided for cooking appliance hazards that involve combustible cooking media (vegetable oils and animal oils and fats).

10.9.3 Portable fire extinguishers shall be provided for other hazards in kitchen areas and shall be selected and installed in accordance with NFPA 10.

NFPA 10

5.5.5. Fire extinguishers provided for the protection of cooking appliance that use combustible cooking media (vegetable or animal oils and fats) shall be listed and labeled for Class K fires.

5.5.5.3 Where a hazard is protected by an automatic fire protection system, a placard shall be conspicuously placed near the extinguisher that states that the fire protection system shall be actuated prior to using the fire extinguisher.

Cooking appliances/processes that do not require the coverage of a fixed fire suppression system.

Cooking appliances tested in accordance with ANSI/UL 197 “Standard for Commercial Electric Cooking Appliances” or equivalent for reduced emissions **and** the volume of grease discharge does not exceed that of normal residential family use do not require the coverage of a fixed fire suppression system. (NFPA 96 – 2017 Sentence 4.1.1.2 specifies the amount of grease discharge that would fall within this scope.)

Specifically, cooking appliances that meet the above criteria include (but is not limited to):

- Steamers
- Rice cookers
- Soup kettles
- Proofers
- Popcorn machines (Note: Kettle corn poppers come in various shapes/sizes and are gas, propane or electrically powered. The authority having jurisdiction (AHJ) has the authority to determine whether these appliances require suppression and ventilation based on the application of the popper).
- Enclosed appliances such as:
    - Ovens, including cook and hold ovens, warming ovens and steam ovens
    - Pizza Ovens (pizza decks)
    - Chicken Rotisseries with doors
    - Tandoori Ovens, Masonry or Cement Pizza Ovens
    - Ventless fryers

Note:

Tandoori Ovens, Masonry or Cement Pizza Ovens have not been tested for fixed fire suppression systems. These appliances are generally constructed of clay or Bakelite material and operate at very high temperatures. Protecting these ovens with a nozzle directly aimed at the oven opening is not recommended due to the explosion hazard this may cause. These ovens must be installed as per the manufacturer’s instructions and the NBC(AE).

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While these cooking appliances do not require the coverage of a fixed fire suppression system, an approved ventilation system is required.

Kitchen fixed fire suppressions systems are required to be engineered or have professional involvement by a registered professional member of the Association of Professional Engineers and Geoscientists of Alberta (APEGA).

Manufacturers may provide documentation to indicate their fixed fire suppression systems consists of pre-engineered and listed components suitable for a specific size, type and location of hazard. Where the manufacturer does not provide engineered documents for the installer's use, documentation indicateing the verification process has been reviewed and authenticated by a registered engineering professional must be provided.

If a fixed fire suppression system has not been fire tested by a Listing Agency, (protection specified in the Manufacturer's Listed Manual), the system is considered a non-engineered system. In such cases, site-specific drawings and the authentication by a registered engineering professional is required.

The owner, fire and building AHJs are required to evaluate the proposed activities, plans and procedures to determine the requirements for fixed fire suppression and exhaust systems.

An overview of the requirements for the installation and maintenance of fixed fire suppression systems and kitchen exhaust systems.

The only acceptable training for the installation of a fixed fire suppression system is the specific training provided by the manufacturer of the system. Companies provide specific training for the installation and maintenance of the systems they manufacture. Manufacturer training for a specific type of equipment is only acceptable for systems made by that manufacturer. Training from other manufacturers on similar systems, or even training from the same manufacturer for a different type of system, is not acceptable training. For example, a person having only suppression system training from manufacturer "x" shall not work on any other type of system made by the manufacturer and must not perform maintenance on any systems from another manufacturer.

Fixed fire suppression systems must be inspected and maintained semi-annually by a qualified person. This requirement is independent from the cleaning frequency of the exhaust/hood. Maintenance includes work such as repair, replacement, or servicing of equipment to ensure it works properly. Cleaning is to remove grease, oil deposits, and other residue.

Outside of manufacturer's training, only the Southern Alberta Institute of Technology (SAIT) offers an approved certificate of training course. Their program material only covers pre-engineered kitchen fixed fire suppression systems. Once a person graduates, the expectation is they gain practical experience while working under the supervision of an experienced, qualified person.

A person qualified to install and/or maintain systems will provide written training certificates to owners, designers and the AHJ as required.

NFC(AE) Division B

2.6.1.9 Commercial Cooking Equipment

6) Commercial cooking equipment that is certified shall be installed and maintained in conformance with its certification.

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7) Uncertified commercial cooking equipment shall be installed and maintained so as not to create a fire hazard.

NFC(AE) Division C

2.2.4.1. Special Fire Suppression Systems
(See Note A-2.2.4.1.)
1) Only qualified persons shall install special fire suppression systems when they have acquired a certificate of training from a manufacturer and on a specific system from the manufacturer.

2) A qualified individual for the installation of a special fire suppression system is a person who has received
a) training from a manufacturer, and
b) training on each specific system that a manufacturer provides before it can be installed.

3) Only qualified persons shall test or perform maintenance on a special fire suppression system when they have acquired an *approved* certificate of training from a public post-secondary educational institution, or they have acquired a certificate of training from a manufacturer for each of their specific systems.

A-2.2.4.1. Proof of training from the manufacturer is provided via an issued certificate with the following information:
(a) certificate holder,
(b) certificate holder company,
(c) date of certification,
(d) type of certification, including the manufacturer and type of system, and
(e) expiry date, if applicable.

Typically, certification must be updated following a change of employment or change of manufacturer. It is the responsibility of the certificate holder to contact the manufacturer regarding any changes in their employment.

Currently, there is only one approved course at a public post-secondary institution, which is provided by the Southern Alberta Institute of Technology (SAIT). The SAIT program material only covers pre-engineered kitchen suppression systems. Once a person graduates, the expectation is that they gain practical experience while working under the supervision of an experienced, qualified person.

Qualified persons for the cleaning commercial cooking equipment (ventilation hoods, fans, ducts and filters).

Cleaning contractors for kitchen exhaust systems must have recognized training and knowledge in cleaning procedures. This training is based on NFC(AE) and NFPA 96 and may include a detailed in-shop training program, policies and procedures for the operation of a business, as well as training obtained from outside sources.

A qualified cleaning contractor will provide owners with a maintenance certificate that must either be attached to the kitchen hood or posted in the kitchen area where it is clearly visible to the fire AHJ. The maintenance certificate must detail the following information:

- Name, address and phone number of cleaning company,
- Printed name and signature of the qualified on-site person supervising the cleaning,

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- Date of cleaning and/or inspection,
- Next cleaning due date and recommended cleaning frequency,
- Indication of what areas were cleaned,
- Notes on why any areas were omitted, and
- Name of the recognized training organization from which the supervisor and/or the cleaning crew members have received their qualification.

Currently, there are only two organizations recognized and approved by the Provincial Fire Administrator for Alberta to provide training for the cleaning of commercial cooking equipment exhaust systems. These organizations are:

1) Phil Ackland Kitchen Exhaust Certification and Training; and

2) MFS Exhaust Hood Cleaning School

Note: To be considered as an approved training provider for kitchen exhaust and hood cleaning an organization must submit their entire curriculum to the Provincial Fire Administrator for review.

NFC(AE) Division B

2.6.1.9 Commercial Cooking Equipment

3) Hoods, grease removal devices, fans, ducts, and other appurtenances shall be cleaned at frequent intervals to prevent surfaces from becoming heavily contaminated with grease or other residues. (See Note A-2.6.1.9.(3).) (See also Article 2.2.4.5. of Division C.)

A-2.6.1.9.(3) Depending on the amount of cooking equipment usage, the entire exhaust system, including grease extractors, should be inspected at intervals not greater than 7 days to determine if grease or other residues have been deposited within. When grease or other residues are in evidence as deposits within the hood, grease removal devices, or ducts, the system should be cleaned. In general, exhaust systems should be cleaned at intervals not greater than 12 months, but in the case of deep fat cooking, char broiling or similar cooking operations, the systems should be cleaned at intervals not greater than 3 months.

NFC(AE) Division C

2.2.4.5. Commercial Cooking Equipment Exhaust Systems
1) Except as provided in Sentence (2), only qualified persons shall perform maintenance on commercial cooking equipment exhaust systems when they have obtained a certificate verifying that they have completed an *approved* course of training in duct-cleaning procedures.
2) A person may clean a commercial cooking equipment exhaust system while under the supervision of a person who has obtained an *approved* certificate of training with the following stipulations:
a) a certified person must be on site for the duration of the cleaning process,
b) a certified person may only have up to two persons “in training” working under them on site, and
c) the person “in training” must be in the process of completing their practicum hours and must be a registered student with a recognized training facility.

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Exhaust and hood cleaning.

The minimum requirement for a qualified person to do exhaust and hood cleaning is annually. However, the AHJ may require an owner to conduct more frequent cleaning based on the accumulation of grease, dust or other residue collected in the hood and duct system.

Changing or moving commercial cooking appliances.

Movement of appliances under the same hood, and the movement or switching of nozzles by a qualified technician, would not require additional engineering, provided there is no addition of new appliances in both number and type.

If this involves changes to energy supplies (new wiring, new piping) a gas and/or electrical permit is required. If the fire alarm connection is new or added, the local AHJ should be contacted as to whether a registered professional member's review of the fire alarm system is required.

Some pre-engineered systems have overlapping appliance protection which means that some appliance positions can be moved as long as they are within the designed overlapping zone. It is important the manufacturer's installation, operation and maintenance manuals are read to ensure that all appliances are protected appropriately.

Note: If wheeled appliances are moved during cleaning operations, the cleaning contractor must ensure all wheeled appliances are placed back in the original designed location once the cleaning has been completed. Appliances, such as wheeled deep fat fryers, are required to be covered by a fixed fire suppression system which is designed and engineered specifically for that cooking appliance in that specific location. Relocating any cooking appliance from the original designed location will require re-engineering to ensure the new location of those wheeled appliances are protected.

Records of cleaning and maintenance must be kept at the business.

Cleaning and maintenance records must be kept on the premise for the AHJ's review. This requirement is based on NFC(AE) Article 2.2.1.2 of Division C and NFPA 96 Sentence 11.2.8.1 .

NFC(AE) Division C

2.2.1.2. Records
3) Records of tests, inspections, maintenance or operational procedures undertaken after the initial tests referred to in Sentence (2) shall be retained so that at least the current and the immediately preceding records are available.
4) Notwithstanding the conditions stated in Sentence (3), no record shall be destroyed within five years of having been prepared and records shall be retained on site.

NFPA 96

11.2.8.1 Records, including certificates of inspection and maintenance, shall be permitted to be forwarded to or shared with the authority having jurisdiction either by hard copy or electronically.

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Requirements when used exhaust hoods or cooking appliances are placed into a facility.

Permits for building, electrical and gas will be required. AHJs have the discretion to accept used equipment if it continues to meet the requirements of codes and standards under the *Safety Codes Act*. As such, fixed fire suppression systems not certified to ANSI/UL 300 and/or ULC/ORD 1254.6 are not acceptable.

Requirements for the interconnection of exhaust, suppression, fuel/energy and fire alarm systems.

The forced draft from hood exhaust fans helps move the fire suppression agent through the ventilation system, thus aiding fire suppression within the duct system. These fans also remove smoke and provide a cooling effect in the plenum and duct after the fire suppressant has been discharged. The system is ULC listed with or without fan operation.

NFPA 96

8.2.3 Exhaust Fan Operation.
8.2.3.1 A hood exhaust fan(s) shall continue to operate after the extinguishing system has been activated unless fan shutdown is required by a listed component of the ventilation system or by the design of the extinguishing system.

Of note, the requirements for kitchen ventilation for relocatable industrial accommodations are specifically addressed in Subsection 10.6.4 of NBC(AE).

NBC(AE)

10.6.4.1. Cooking Equipment Ventilation System
1) Except as provided in Article 10.6.4.2., every kitchen containing commercial cooking equipment used in processes producing smoke or grease-laden vapours shall be equipped with a mechanical exhaust system conforming to NFPA 96, “Ventilation Control and Fire Protection of Commercial Cooking Operations.”

10.6.4.2. Kitchen Hoods, Canopies and Exposed Exhaust Ducts
1) Ducts for a kitchen exhaust system shall be constructed of 0.84 mm minimum thickness stainless steel.
2) A demountable exhaust extension may be used if the connection is exposed and is grease-tight.
3) The airflow in and around a canopy or hood shall be in accordance with good engineering practice and each design shall be submitted to the *authority having jurisdiction* for review.
4) The required clearance from the *exhaust duct* to *combustible* material may be waived if a 25 mm air space, having no materials in it, separates the *exhaust duct* from a *noncombustible* material backed by not less than 25 mm of mineral wool insulation which protects the *combustible* material.
5) The required clearance from the hood or canopy to *combustible* material may be waived if a 50 mm air space, having no materials in it, separates the hood from a *noncombustible* material backed by not less than 25 mm of mineral wool insulation which protects the *combustible* material.
6) A sidewall fan may be used.
7) A fan shall be rated for continuous use as a commercial exhaust fan.

Requirements for simultaneous operation of multiple hoods in food courts, new teaching and/or multi-use kitchens in new or recently renovated Alberta schools.

The requirement for simultaneous operation of the fixed fire suppression system is to define the hazard area. A single hazard area can be defined as multiple hoods installed end to end, back-to-back, or both and is less than

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75ft (22.9m) in length and shares an interconnected (common) duct and have a grease-producing appliance located under one or more of the hoods. All the fixed fire suppression systems are required to activate upon the actuation of any one of these systems.

If the installation of hoods exceeds 75ft (22.9m) in length, then the ductwork beyond the 75ft must be protected by an independent fixed fire suppression system with its own detection system or by a fixed fire suppression system whichactivates simultaneously with the fixed fire suppression systems protecting the hoods.
Notes:

- Both NFPA 17A and 96 provide information on the Simultaneous Operation and the use of interconnecting of exhaust ducts. These documents should be referred to for further information.
- Significant changes on simultaneous operation have changed from previous referenced editions of NFPA 96. NFPA 96-2017 is the referenced edition in both NBC(AE) and NFC(AE).

NFPA 96

10.3 Simultaneous Operation.
10.3.1 Fixed pipe extinguishing systems in a single hazard area (see 3.3.44 for the definition of single hazard area) shall be arranged for simultaneous automatic operation upon actuation of any one of the systems.
10.3.1.1 Hoods installed end to end, back to back, or both, or sharing a common ductwork, not exceeding 22.9 m (75 ft) in distance from the farthest hood, and having a grease-producing appliance(s) located under one or more of the hoods, shall be considered a single hazard area requiring simultaneous automatic fire protection in all hoods and ducts.
10.3.1.1.1 In hoods that are installed end to end, back to back, or both, and that share a common ductwork, the ductwork beyond 22.9 m (75 ft) from the farthest hood shall be protected by an independent fire-extinguishing system with its own detection system or by a fire-extinguishing system that activates simultaneously with the fire-extinguishing system(s) protecting the hoods.
10.3.1.2 Hoods installed end to end, back to back, or both that do not share a common exhaust duct and are separated by a wall(s) or other means to ensure that grease-laden vapors exhausted under one hood cannot propagate to the other hoods, the hoods' fire-extinguishing system(s) shall be independent and shall not be required to simultaneously discharge.

This Interpretation replaces 19-FCI-002R1 Fixed fire suppression and exhaust systems – cooking, cleaning and maintenance.

This INTERPRETATION is applicable throughout the Province of Alberta.

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Unless stated otherwise, all Code references in this STANDATA are to Division B of the National Fire Code – 2023 Alberta Edition and National Building Code – 2023 Alberta Edition.

Issued by the Provincial Fire and Building Administrators

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